Dynamical Localization of Quantum Walks in the regime of strong disorder

QMath16 - Random Systems

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02/09/2025

Co-funded by the European Union

Quantum Walks on the hexagonal lattice

Intuition: Quantum particle moving in discrete time on discrete grid Γ.
State of quantum particle with internal degree of freedom s is given by $\psi \in l^2(\Gamma) \otimes \mathbb{C}^s$.

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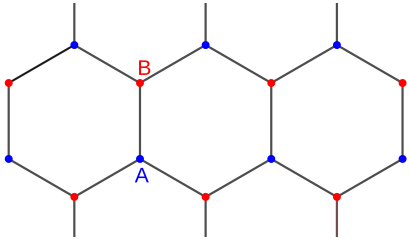
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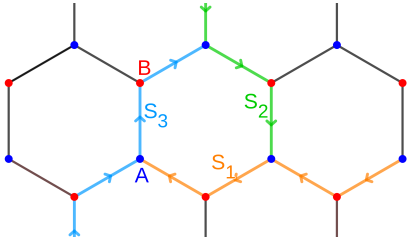
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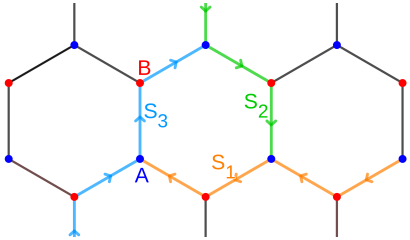
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$\Longrightarrow$ Quantum Walk: $U = \mathcal{SC}$



Transport Properties

Translation-invariant case:

Fourier Transform: $\widehat{U} = \mathcal{F} U \mathcal{F}^{-1}$ is multiplication operator on $L^2(\mathbb{T}^2, \mathbb{C}^6)$

$\rightsquigarrow \sigma_{\text{cont}}(U) = \varnothing \iff k \in \mathbb{T}^2 \mapsto \text{tr}\Big(\widehat{U}(k)\Big) = \text{const.}$

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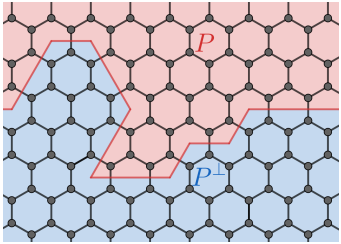
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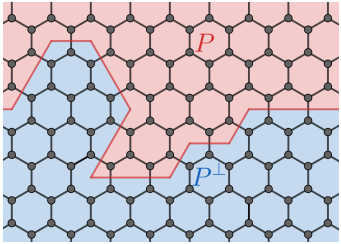
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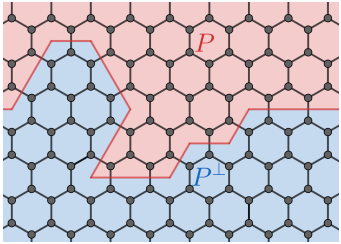
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$\text{ind}(U) \neq 0 \Rightarrow$ criteria on $\sigma(U), \sigma_{\text{ac}}(U) = \mathbb{S}^1$, e.g. $|(C_x)_{2,1}|, |(C_x)_{1,2}|, |(C_x)_{3,3}| \to 1$ ($|x| \to \infty$ on boundary), then $\sigma(U) = \mathbb{S}^1$



Dynamical Localization

Equip every site $x \in \Gamma$ with i.i.d. RV ω_x uniformly distributed on $[0, 2\pi]$
$\implies$ define $D_\omega(|x\rangle \otimes e_i) = e^{i\omega_x} |x\rangle \otimes e_i$ and $U_\omega = D_\omega U$

Definition

Dynamical localization: $\exists\, g > 0$ such that $\forall\, x, y \in \Gamma$:

$$\mathbb{E}\Big(\sup_{n\in\mathbb{Z}} |\langle x| U_\omega^n |y\rangle| \Big) \leqslant c\, e^{-g|x-y|}$$

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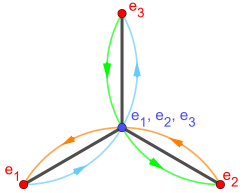
Dyn. Loc. implies spectral localization, i.e. U_ω has pure point spectrum a.s.
Fractional moments criterion: for some $s \in (0,1)$ and $1 - \epsilon < |z| < 1$

$$\mathbb{E}\Big(|\langle x| (U_\omega - z)^{-1} |y\rangle|^s \Big) \leqslant c\, e^{-g|x-y|}$$

The result

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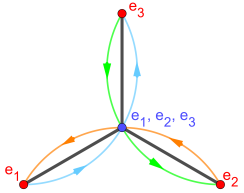
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Theorem 1

For any $s \in (0, \frac{1}{3})$ there exists $g, \varphi, c > 0$ such that for any $\{C_x\}_{x\in\Gamma} \subset U(3)$ with $\sup_x \|C_x - C_0\| \leqslant \varphi$ and all $x, y \in \Gamma$, $z \in \mathbb{C}$ with $|z| < 1$:

$$\mathbb{E}\left(|\langle x \otimes e_i| \, (U_\omega - z)^{-1} \, |y \otimes e_j\rangle|^s\right) \leqslant c \, e^{-g|x-y|}.$$

Proof: Finite volume method

Correlated phases

Recall $D_\omega(|x\rangle \otimes e_i) = e^{i\omega_x}\ |x\rangle \otimes e_i$, we write D_ω in the ONB $\big(|x\rangle \otimes e_i\big)_{x\in\Gamma, i=1,2,3}$:

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Problem: "Fractional Moment Criterion $\Rightarrow$ Dynamical Localization" requires independent phases

Thank you for your attention!

Reference:

Dynamical Localization and Transport properties of Quantum Walks on the hexagonal lattice, arXiv:2503.09421

Finite volume criterion

Restrict QW U_ω to box Λ_L of size $L \in^2$

Iterative step:

$\exists\, q < 1$ s.t. for $s < \frac{1}{3}$, L big enough $\exists\, \varphi > 0$:

$$\mathbb{E}\left(\left|\langle x^{(i)}\right| (U_\omega - z)^{-1} \left|y^{(j)}\right\rangle\right|^s\Big) \leqslant q \max_{v^{(k)} \in \partial(\Lambda_L + x)} \mathbb{E}\left(\left|\langle v^{(k)}\right| (U_\omega - z)^{-1} \left|y^{(j)}\right\rangle\right|^s\Big)$$

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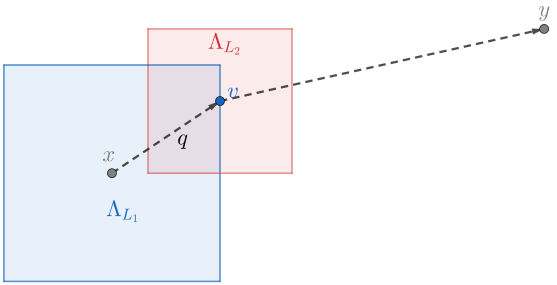
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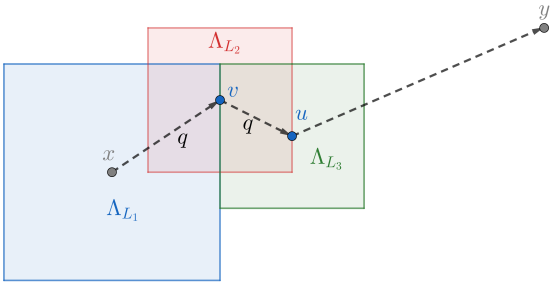
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